

# Persistent Topological Divergence for Provable Hallucination Detection

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# The Hallucination Problem

## What LLMs Do Well

- Generate fluent text
- Answer complex questions
- Perform reasoning tasks

## The Critical Flaw

- Generate plausible but **false** information
- Create internally consistent **fabrications**
- Undermine trust in high-stakes domains

**How do we detect when LLMs hallucinate?**

# Existing Detection Methods

| Paradigm                                     | Core Idea                                  | Key Limitation                 |
|----------------------------------------------|--------------------------------------------|--------------------------------|
| Knowledge-Grounded                           | Check facts against external databases     | Latency, coverage, cost        |
| Self-Consistency                             | Generate multiple outputs, check agreement | High computational cost        |
| Model-Internal                               | Analyze hidden states                      | Requires training data         |
| structure.fgl10<br><b>TOHA (Topological)</b> | Measure connectivity to prompt             | <b>Misses complex patterns</b> |

**Problem:** Existing topological methods only measure *connectivity*. They miss higher-order structural inconsistencies.

**TOHA's approach:** Measure how "connected" response tokens are to the prompt

## Failure Case: "Connected but Incoherent"

A hallucinated response can:

- Have strong token-by-token attention to the prompt
- Form an internally consistent but **factually incorrect** narrative loop
- Score **low** on TOHA (looks well-connected)

## Why this happens:

- TOHA only uses  $H_0$  (connectivity, connected components)
- Blind to  $H_1$  (loops),  $H_2$  (voids), higher structures
- Measures minimum spanning forest length with linear error in response length

# Persistent **T**opological **D**ivergence for **E**valuation

## Key Idea: Dual Topological Views

A faithful response must be:

- ① **Internally coherent** (has its own structure)
- ② **Grounded to the prompt** (anchored to context)

**Hallucination** = Large mismatch between these two views

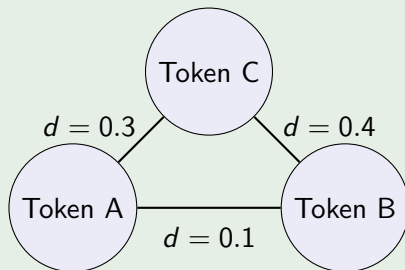
# From Attention to Topology

**Step 1:** LLM produces attention matrix  $W = [w_{ij}]$  where  $w_{ij} \in [0, 1]$

**Step 2:** Convert to pseudo-distances:  $d(i, j) = 1 - w_{ij}$

- Strong attention ( $w_{ij} \approx 1$ )  $\rightarrow$  small distance ( $d \approx 0$ )
- Tokens are "topologically close"

## Example



# Persistent Homology: The Key Tool

**Idea:** Track how topology changes as we vary a scale parameter

## Vietoris-Rips Filtration

At scale  $\epsilon$ , connect tokens  $i$  and  $j$  if  $d(i, j) \leq \epsilon$

As  $\epsilon$  increases: isolated points  $\rightarrow$  clusters  $\rightarrow$  loops  $\rightarrow$  higher structures

• • • •  
 $\epsilon = 0$   
4 components

— • — • •  
 $\epsilon = 0.5$   
2 components

— • — • — •  
 $\epsilon = 1.0$   
1 component

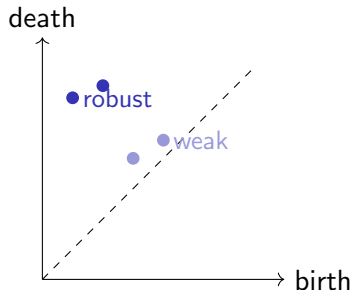
**Output:** Persistence diagram = birth/death times of topological features

## What They Encode

Each point  $(b, d)$  represents:

- A topological feature
- Born at scale  $b$
- Dies at scale  $d$
- Persistence =  $d - b$

Points far from diagonal = robust features



We compare diagrams using **Wasserstein distance**: optimal matching cost between features



# The Two Topologies

## 1. Internal Response Topology

**Question:** What is the response's own narrative structure?

**Construction:**

- Take only response tokens  $V_R$
- Use attention weights between response tokens:  $W_{RR}$
- Build Vietoris-Rips filtration  $\rightarrow$  persistence diagram  $PD_k(G_R^{\text{internal}})$

## 2. Grounded Response Topology

**Question:** How is each part anchored to the prompt?

**Construction:**

- For each response token  $r$ , define grounding:  $f_{\text{ground}}(r) = \min_{p \in V_P} (1 - w_{rp})$
- Small value = strongly connected to prompt
- Build sublevel set filtration  $\rightarrow$  persistence diagram  $PD_k(G_R^{\text{grounded}})$

# The PerToDiVE Score

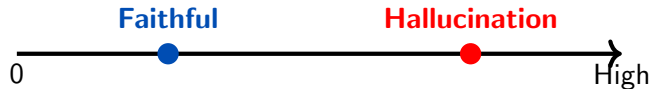
## Formal Definition

$$\text{PerToDiVE}_k(P, R) = W_p \left( \text{PD}_k(G_R^{\text{internal}}), \text{PD}_k(G_R^{\text{grounded}}) \right)$$

where  $W_p$  is the  $p$ -Wasserstein distance (typically  $p = 2$ )

## Interpretation:

- **Low score:** Internal structure matches grounding → likely faithful
- **High score:** Large mismatch → potential hallucination



# Visual Example: Internal vs Grounded

## Faithful Response



Internal: linear

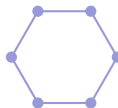


Grounded: linear

**Similar diagrams**

**Low PerToDiVE**

## Hallucinated Response



Internal: loop ( $H_1$ )



Grounded: disconnected

**Very different!**

**High PerToDiVE**

# Theoretical Guarantees: Stability

**Question:** Is PerToDiVE robust to small numerical perturbations?

## Lipschitz Stability Theorem

If attention weights change by at most  $\delta$ , then:

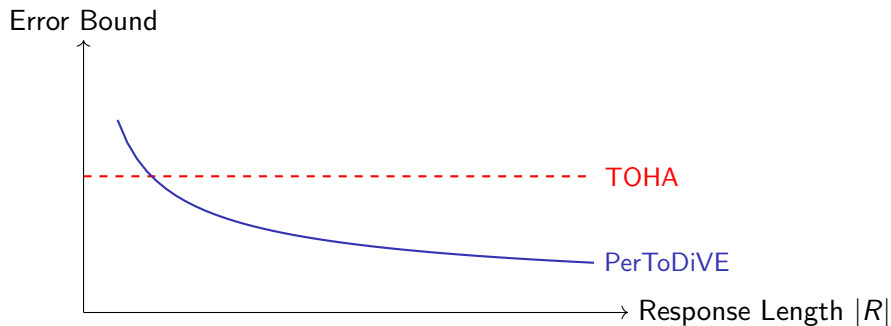
$$|\Delta \text{PerToDiVE}_k| \leq C_k \cdot \delta$$

where  $C_k = |V_R|^{1/p}$  for the  $p$ -Wasserstein distance.

| Method           | Raw Error Bound          | Per-Token Error            |
|------------------|--------------------------|----------------------------|
| TOHA             | $ R  \cdot \delta$       | $\delta$ (constant)        |
| <b>PerToDiVE</b> | $ R ^{1/p} \cdot \delta$ | $ R ^{1/p-1} \cdot \delta$ |

**Key insight:** For  $p > 1$ , our error *decreases* as responses get longer!

# Why Sub-Linear Scaling Matters



For long, complex generations

**PerToDiVE is provably more  
reliable for long-form text**

Critical for detailed explanations, reports, and reasoning

# Higher-Dimensional Features Matter

**TOHA limitation:** Only uses  $H_0$  (connectivity)  
 $H_k$  for all dimensions

**PerToDiVE advantage:** Captures

## What Different Dimensions Detect

- $H_0$ : Connected components     $H_1$ : Loops (circular reasoning)
- $H_2$ : Voids (gaps in reasoning)    Higher  $H_k$ : Multi-scale inconsistencies

## Example: Circular Reasoning

A hallucinated response creates a self-reinforcing loop of claims grounded in each other but not the prompt:

- High  $H_1$  persistence in internal topology
- Low  $H_1$  persistence in grounded topology
- Large Wasserstein distance  $\rightarrow$  detected!

# Formalizing "Ungrounded" Responses

## Definition: $(k, \tau)$ -Topologically Ungrounded

A response  $R$  is  $(k, \tau)$ -ungrounded if:

- Its internal topology has significant features with persistence  $\geq \tau$
- Its grounded topology has mostly features with persistence  $< \tau$

## Proposition

If  $R$  is  $(k, \tau)$ -ungrounded for large  $\tau$ , then  $\text{PerToDiVE}_k(P, R)$  is large.

**Intuition:** The Wasserstein distance must pay a high cost to match persistent internal features to weak/absent grounded features.

This formalizes the connection: structural mismatch  $\Rightarrow$  high score  $\Rightarrow$  hallucination signal

## Algorithm overview:

- 1 Extract attention matrix  $W$  from LLM
- 2 Compute internal persistence diagram (VR filtration on  $V_R$ )
- 3 Compute grounded persistence diagram (sublevel filtration)
- 4 Calculate Wasserstein distance between diagrams

## Complexity Analysis

- Persistence computation:  $O(n^3)$  worst case, faster in practice
- Wasserstein distance: Near-linear with kd-tree auction algorithm
- **Total:** Practical for real-time detection

Single forward pass required (unlike self-consistency methods). Can be computed for selected attention heads only.



## Why PerToDiVE is Better

**vs. TOHA:** Captures higher-dimensional features ( $H_1, H_2, \dots$ ), not just connectivity; sub-linear error scaling instead of linear

**vs. Self-Consistency:** Single forward pass (not multiple generations); no sampling overhead

**vs. Knowledge-Grounded:** No external database required; works for proprietary or specialized domains

**Provable Guarantee:** First multi-scale topological detector with formal Lipschitz stability theorem

# Comparison Table

| Property        | TOHA       | SelfCheck | Knowledge | PerToDiVE  |
|-----------------|------------|-----------|-----------|------------|
| External data   | No         | No        | Yes       | No         |
| Multiple passes | No         | Yes       | Varies    | No         |
| Training data   | No         | No        | Yes       | No         |
| Dimensions      | $H_0$ only | N/A       | N/A       | All $H_k$  |
| Stability bound | Linear     | None      | None      | Sub-linear |
| Detects loops   | No         | Limited   | N/A       | Yes        |
| Formal proof    | Weak       | None      | None      | Strong     |

PerToDiVE combines the best properties: efficient, model-internal, and provably robust

# Key Contributions

**Dual-  
Filtration  
Framework**

Compare internal vs  
grounded structure

**Multi-Dim.  
Topology**

Capture loops,  
voids, not just  
connectivity

**Provable  
Stability**

Sub-linear error  
scaling with length

**First topological detector with  
formal multi-scale guarantees**

## Immediate Next Steps

- **Empirical validation:** Test on benchmark datasets (TruthfulQA, HaluEval)
- **Optimize computation:** Implement efficient persistence algorithms
- **Head selection:** Develop principled methods for identifying informative attention heads

## Longer-Term Research

- Extend to other model architectures (non-Transformer models)
- Explore connections to causal interventions in attention
- Develop real-time monitoring systems for deployed LLMs
- Investigate topological signatures of different hallucination types

# Broader Impact

## Scientific

- Bridges TDA and NLP communities
- Opens new research directions in interpretable AI
- Provides toolkit for analyzing attention mechanisms

## Practical

- Enables safer LLM deployment
- Applicable to high-stakes domains (medicine, law)
- Framework-agnostic approach

Moving from *heuristic detection* to *provable reliability*

## The PerToDiVE Framework

- **Problem:** LLM hallucinations undermine reliability; existing detectors are incomplete
- **Solution:** Compare internal response topology with prompt-grounded topology using persistent homology
- **Innovation:** First multi-dimensional topological detector with provable sub-linear stability
- **Impact:** Moves field toward principled, theoretically grounded detection methods

## Theoretical

- Can we tighten the stability bound further?
- What is the information-theoretic limit of topology-based detection?
- How do different persistence dimensions correlate with hallucination types?

## Practical

- Which attention heads are most informative across different models?
- How does performance scale to very long contexts (100K+ tokens)?
- Can we adapt this framework to multimodal models?

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